(Example: *Chlorobium*) and green non sulphur bacteria (Example: *Chloroflexus*) purple sulphur bacteria (Example: *Chromatium*) and purple non sulphur bacteria (Example: *Rhodospirillum*). These photosynthetic bacteria can fix atmospheric CO_2 in a similar fashion like cyanobacteria but using only one photosystem and using H_2S as the electron donor instead of H_2O.

Process of photosynthesis in bacteria

The electron transport system in purple and green bacteria consists of only one Photosystem PSI (P870). They do not possess photosystem II. When P870 gets excited upon capture of light energy, it donates the electron to bacteriopheophytin. Electrons flow through quinones and cytochromes and are reverted back to P870. This process is cyclic (since the electron excited from P870 comes back to P870) and generates ATP. A reversed electron flow operates in purple bacteria to reduce NAD^+ to NADH. Electrons are extracted from external electron donors like hydrogen sulphide, hydrogen, elemental sulphur and organic compounds to synthesise NADH. Since H_2O is not used as electron donor, oxygen is not evolved which explains the anoxygenic nature of the organisms involved (Figure 6.3). The sulphur evolved during this reaction is deposited as sulphur globules either outside or inside the cells. $CO_2 + H_2S \rightarrow (CH_2O)_n + S$. Table 6.3 compares the photosystheic process in plants, algae and bacteria.

HOTS

1. What will be the electron flow sequence of noncyclic and cyclic photo phosphorylation?
2. Chemical energy produced in photosynthesis is either ATP NADPH or ATP NADH. Why?

6.5 Microbial Growth

In bacteria, growth can be defined as an increase in cellular constituents. Growth results in increase of cell number.

When bacteria are cultivated in liquid medium and are grown as batch culture (Growth occurring in a single batch of medium with no fresh medium provided), cell multiplication happens till all the nutrients are exhausted. After sometime, nutrient concentrations decline and bacterial cells begin to die. This growth pattern can be plotted in a graph as the logarithm of viable cells versus incubation time (Figure 6.4). The growth curve has four distinct phases.

1. Lag phase
2. Logrithmic phase/Exponential phase
3. Stationary phase
4. Death phase

1. Lag Phase

When bacteria are introduced into fresh medium, no immediate cell multiplication and increase in cell numbers occur. The cell prepares itself for cell division by synthesizing cell components and increase in cell mass. Since there is a lag in cell division, this phase is called lag phase.

2. Logrithmic Phase/Exponential Phase

During this phase, microorganisms rapidly divide and grow at a maximal rate possible utilizing all the nutrients present in the medium. The growth rate is constant during the exponential phase. The organism divides and doubles in number at regular intervals. The growth curve rises smoothly.

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